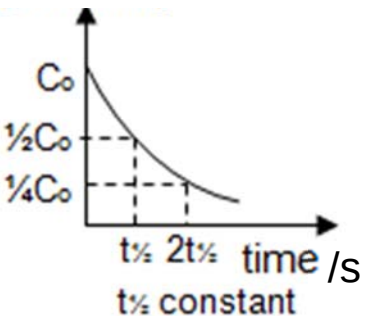


Qn	Suggested Mark Scheme	Mark							
1(a)(i), (ii)	Correctly calculates the <u>volume of FA 1 used</u> and volume of FA 1 used is <u>within 40.00 to 45.00 cm³</u> , both inclusive.	1							
	Tabulates <u>initial</u> and <u>final burette readings</u> for table and <u>volume of FA 2 used</u> in (a)(ii). Table has <u>appropriate headers</u> and <u>units</u> . Tabulation may be <u>vertical or horizontal</u> ; lines are not essential but there should be no absences of headers. Where units have not been included in the header, there should be the <u>appropriate unit</u> for each entry in the table. Do NOT award this mark if any final and initial burette readings in either table are inverted or 50 is used as the initial burette reading in either table.	1							
	All the final/initial <u>burette readings</u> for <u>all accurate titres</u> in <u>BOTH</u> tables are recorded to the <u>nearest 0.05 cm³</u> . <i>Treat all titres as "accurate" unless labelled rough or 1st titre is of lower precision than subsequent titres</i>	1							
	Has at least <u>TWO</u> uncorrected titres for end-point within 0.10 cm ³ * refers to uncorrected titres i.e. ignore arithmetical errors made when calculating the titre values	1							
	Check and correct, if necessary, subtractions in the titre table. Examiner then selects the "best" titre using the hierarchy: two identical; titres within 0.05 cm ³ ; titres within 0.1 cm ³ , etc Calculate student titre $\times \frac{\text{student volume of FA1 added}}{\text{teacher volume of FA1 added}}$ <i>Teacher's results</i> <table border="1"> <thead> <tr> <th>Shift</th><th>volume of FA 1 added / cm³</th><th>average titre volume / cm³</th></tr> </thead> <tbody> <tr> <td>1</td><td rowspan="2">40.90</td><td>25.78</td></tr> <tr> <td>3</td><td>25.85</td></tr> </tbody> </table> Calculate difference in teacher and student scaled values (Δ titre) and award "quality" marks as below. Give 3 marks if this difference is within 0.20 cm³ Give 2 marks if this difference is > 0.20 cm³ but $\leq$ 0.40 cm³ Give 1 mark if this difference is > 0.40 cm³ but $\leq$ 0.60 cm³ Give 0 marks for a difference that is > 0.60 cm³.	Shift	volume of FA 1 added / cm ³	average titre volume / cm ³	1	40.90	25.78	3	25.85
Shift	volume of FA 1 added / cm ³	average titre volume / cm ³							
1	40.90	25.78							
3		25.85							

(a)(iii)	Obtains appropriate 'average', to 2 d.p., from any experiments with uncorrected end-point values within 0.10 cm ³ . <i>Do not award this mark if the titres used are not identified either in the table (by, for examples, a tick) or in a calculation.</i> <i>Do not award this mark if there are arithmetic errors in the table.</i>	1
(b)(i)	Let the volume of FA 1 used for dilution be V_{FA1} cm ³ . $n(\text{MnO}_4^-)$ in FA 1 used = $\frac{V_{FA1}}{1000} \times 0.100 = V_{FA1} \times 10^{-4}$ mol $n(\text{MnO}_4^-)$ in 25.0 cm ³ of FA 5 = $\frac{V_{FA1} \times 10^{-4}}{10} = V_{FA1} \times 10^{-5}$ mol	1 1
(b)(ii)	$n(\text{S}_2\text{O}_3^{2-}) = 5 \times n(\text{MnO}_4^-)$ in 25.0 cm ³ of FA 5 = $V_{FA1} \times 10^{-5} \times 5$ mol	1
(b)(iii)	Let the average volume of FA 2 used = V_m cm ³ . $[\text{S}_2\text{O}_3^{2-}]$ in FA 2 = $V_{FA1} \times 10^{-5} \times 5 \times \frac{1000}{V_m}$ mol dm ⁻³	1
(c)	No, using burettes to measure the volume of FA 3 and FA 4 will not improve the accuracy of the titration data as <u>both FA 3 and FA 4 are used in excess</u> OR the <u>amount of I_2 formed</u> only <u>depends on the amount of KMnO_4 used</u> , which was already accurately measured.	1
(d)	% error in volumetric flask = $\frac{0.15 \times 100}{250} = 0.06\%$ % error in pipette = $\frac{0.1 \times 100}{25.0} = 0.4\%$ % error in burette = $\frac{2(0.05) \times 100}{25.65} = 0.390\%$ Total error = <u>0.850%</u> Also accept if the error for measuring 40.00 cm ³ and 45.00 cm ³ of FA 1 with burette is included, then the total error = 1.07 – 1.1%	1
(e)(i)	Order of reaction = 1	1
(e)(ii)	Since total volume of reaction mixture is kept constant, • $[\text{KI}] \propto V_{\text{KI}}$ used Also since rate $\propto \frac{1}{t}$ (as the volume of $\text{Na}_2\text{S}_2\text{O}_3$ is kept constant) AND • when the volume of KI is doubled, the time taken for the iodine to appear is halved. Rate of reaction is directly proportional to $[\text{KI}]$. OR • $V_{\text{KI}} \times t = \text{constant}$	1 1

(e)(iii)	$[KI] / \text{mol dm}^{-3}$ 	1												
(e)(iv)	- Using a 50 cm³ measuring cylinder, measure 40 cm³ of KI and pour the solution into a 250 cm³ conical flask. - Using a 10 cm³ measuring cylinder, measure 10 cm³ of Na₂S₂O₃ and add this into the same conical flask. - Using a 25 cm³ measuring cylinder, measure 25 cm³ of H₂O and pour into the conical flask. - Measure 25 cm³ of X using another 25 cm³ measuring cylinder. - Rapidly pour X into the conical flask and start the stop-watch at the same time. - Swirl the reaction mixture. - Note the time taken for brown colour of iodine to first appear. <table border="1" data-bbox="279 1019 1372 1825"> <thead> <tr> <th>Marks</th><th>Requirements</th><th>Key Marking Points</th></tr> </thead> <tbody> <tr> <td>[2]</td><td>Appropriate <u>quantities</u> of substances used</td><td> - vol of Na₂S₂O₃ = <u>10 cm³</u> - vol of KI = <u>40 cm³</u> or <u>20 cm³</u> - vol of X = <u>25 cm³</u> (or other volume except 50 cm³) - vol of water added to maintain <u>total volume at 100 cm³</u> (1 mark for any 2 points, 2 marks for all 4 points) </td></tr> <tr> <td>[1]</td><td>Appropriate <u>apparatus</u> and their <u>capacities</u></td><td> - Na₂S₂O₃ : 10 cm³ measuring cylinder or 50 cm³ burette or 10 cm³ pipette - KI / X / H₂O : 50 cm³ / 25 cm³ measuring cylinder or 50 cm³ burette or 25 cm³ pipette 250 cm³ conical flask </td></tr> <tr> <td>[2]</td><td>Essential <u>experimental details</u> to ensure <u>reliable results</u></td><td> - Start reaction by <u>adding X last</u> and <u>rapidly</u> <u>Simultaneously</u> start reaction with starting stopwatch <u>Swirl</u> the reaction mixture <u>Stop stopwatch</u> when <u>brown</u> colour of iodine appears (1 mark for any 2 points, 2 marks for all 4 points) </td></tr> </tbody> </table>	Marks	Requirements	Key Marking Points	[2]	Appropriate <u>quantities</u> of substances used	- vol of Na₂S₂O₃ = <u>10 cm³</u> - vol of KI = <u>40 cm³</u> or <u>20 cm³</u> - vol of X = <u>25 cm³</u> (or other volume except 50 cm³) - vol of water added to maintain <u>total volume at 100 cm³</u> (1 mark for any 2 points, 2 marks for all 4 points)	[1]	Appropriate <u>apparatus</u> and their <u>capacities</u>	- Na₂S₂O₃ : 10 cm³ measuring cylinder or 50 cm³ burette or 10 cm³ pipette - KI / X / H₂O : 50 cm³ / 25 cm³ measuring cylinder or 50 cm³ burette or 25 cm³ pipette 250 cm³ conical flask	[2]	Essential <u>experimental details</u> to ensure <u>reliable results</u>	- Start reaction by <u>adding X last</u> and <u>rapidly</u> <u>Simultaneously</u> start reaction with starting stopwatch <u>Swirl</u> the reaction mixture <u>Stop stopwatch</u> when <u>brown</u> colour of iodine appears (1 mark for any 2 points, 2 marks for all 4 points)	5
Marks	Requirements	Key Marking Points												
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[1]	Appropriate <u>apparatus</u> and their <u>capacities</u>	- Na₂S₂O₃ : 10 cm³ measuring cylinder or 50 cm³ burette or 10 cm³ pipette - KI / X / H₂O : 50 cm³ / 25 cm³ measuring cylinder or 50 cm³ burette or 25 cm³ pipette 250 cm³ conical flask												
[2]	Essential <u>experimental details</u> to ensure <u>reliable results</u>	- Start reaction by <u>adding X last</u> and <u>rapidly</u> <u>Simultaneously</u> start reaction with starting stopwatch <u>Swirl</u> the reaction mixture <u>Stop stopwatch</u> when <u>brown</u> colour of iodine appears (1 mark for any 2 points, 2 marks for all 4 points)												

Qn	Suggested Mark Scheme		Mark								
2	Mass of empty weighing bottle / g		1								
	Mass of weighing bottle and FA 7/8 / g										
	Mass of weighing bottle and residual FA 7/8 / g										
	Mass of FA 7/8 used / g										
	Highest / Lowest temperature reached / °C										
	Initial temperature of FA 6 / °C										
	Temperature rise / drop / °C										
	Or Change in temperature / °C										
	• Tabulates <u>weighings</u> / <u>temperature</u> in appropriate manner. Tabulation may be vertical or horizontal; lines are not essential but there should be no absence of headers. • Correct use of <u>highest</u> and <u>lowest</u> temperature. • Correct use of <u>rise/ drop</u>. • Tables have correct <u>headers</u> and units. Where <u>units</u> have not been included in the header, there should be the appropriate unit for <u>each entry</u> in the tables. • For students who record “Change” in temperature, the corresponding sign must be included. • <i>Ignore the absence of a calculated mass for FA 7/8.</i>										
Complete set of masses (refer to table above) <i>Do not award mark if Tare function is used.</i> The mass of FA 7/8 weighed is within specified limits. 4.45 g ≤ FA 7 ≤ 5.05 g 3.45 g ≤ FA 8 ≤ 4.05 g Correctly calculate the mass of FA 7/8 .		1									
All the mass readings are recorded to 3 d.p. All temperatures are recorded to 0.1 °C		1									
<u>Accuracy mark.</u> If $\frac{\Delta T}{\text{mass of FA 7/8}}$ for both experiments +/- 0.05 from Teacher's result If $\frac{\Delta T}{\text{mass of FA 7/8}}$ for both experiments +/- 0.2 from Teacher's result <i>Teacher's results</i> <table><tr><th>Shift</th><th>ΔT / m for FA 7</th><th>ΔT / m for FA 8</th></tr><tr><td>1</td><td>6.2 / 4.523 = 1.371</td><td>5.4 / 3.620 = 1.492</td></tr><tr><td>3</td><td>6.4 / 4.831 = 1.325</td><td>5.6 / 3.674 = 1.524</td></tr></table>		Shift	ΔT / m for FA 7	ΔT / m for FA 8	1	6.2 / 4.523 = 1.371	5.4 / 3.620 = 1.492	3	6.4 / 4.831 = 1.325	5.6 / 3.674 = 1.524	2 (max)
Shift	ΔT / m for FA 7	ΔT / m for FA 8									
1	6.2 / 4.523 = 1.371	5.4 / 3.620 = 1.492									
3	6.4 / 4.831 = 1.325	5.6 / 3.674 = 1.524									

(b)	Correctly calculates no of moles of $\text{HCl} = 1 \times \frac{50}{1000} = 0.0500 \text{ mol}$	1
(c)	Correctly calculates number of moles of Na_2CO_3 used = $\frac{\text{mass of FA 7 used}}{106.0}$	1
(d)	Since <u>0.0500 moles</u> of HCl requires only <u>0.0250 moles</u> of Na_2CO_3 for complete reaction, <u>Na_2CO_3</u> is in excess.	1
(e)	Correctly calculates $\Delta H_1 = -\frac{q}{n_{\text{HCl}}} = -\frac{q}{0.0500} \times 2 = \underline{-a} \text{ kJ mol}^{-1}$ Correct sign for $\Delta H_1 = \text{-ve}$	1
(f)	Correctly calculates $q = V_{\text{FA 6}} c \Delta T = 50 \times 4.2 \times \Delta T$ Ignore if q has sign or has a wrong unit. Ignore if ΔT was calculated wrongly.	1
(g)	Correctly calculates number of moles of NaHCO_3 used = $\frac{\text{mass of FA 8 used}}{84.0}$	1
(h)	Recognises that the limiting reagent is NaHCO_3 and uses the number of moles of NaHCO_3 to determine ΔH . Correctly calculates $\Delta H_2 = +\frac{q}{n_{\text{NaHCO}_3}} = \underline{+b} \text{ kJ mol}^{-1}$ Correct sign for $\Delta H_2 = \text{+ve}$	1
(i)	Correctly calculates $\Delta H_3 = 2\Delta H_2 - \Delta H_1 = \underline{2b - (-a)} \text{ kJ mol}^{-1}$	1
	Shows working in all calculations in 1(a)(iii), 1(b), 2(a) – (c) and 2(e) – (i) . All calculations must be relevant although they may not be complete or correct. <i>Any calculations not attempted loses this mark.</i>	1
	Shows appropriate significant figures (3 s.f.) in all final answers in 1(b), 2(a) – (c) and 2(e) – (i) . Shows appropriate units in 1(a)(iii), 1(b), 2(a) – (c) and 2(e) – (i) . <i>Any calculations not attempted loses this mark.</i>	1
(j)	<u>No effect.</u> This is because <u>HCl</u> was already used <u>in excess</u> , hence ΔH_2 calculated is <u>independent</u> of the number of moles of <u>HCl</u> . Therefore, <u>ΔH_3</u> will not be affected.	1

Qn	Suggested Mark Scheme		
3(a)		observations for FA 9	observations for FA 10
	Add NaOH(aq), followed by a spatula load of aluminium powder.	blue ppt (1), insoluble in excess NaOH (1). pungent gas evolved turned moist red litmus blue (1) – NH ₃ (1)	
	Add concentrated HCl dropwise.	blue solution turns (green) then yellowish green/yellow (1) no effervescence observed (1). no NO ₂ gas evolved (1).	
	To the FA 9 and concentrated HCl mixture, add deionised water.	yellowish–green/yellow solution turns (green) then blue (1)	
	Add KI(aq).	(white/cream) ppt in brown solution (1)	
	To the resulting solution, add Na ₂ S ₂ O ₃ (aq).	brown colour discharged (1) white/cream ppt observed (1)	
	Add 1 cm ³ dilute H ₂ SO ₄ followed by 1 cm ³ of K ₂ CrO ₄ (aq).		pale blue green solution turns (green) then yellowish–brown (1)
	Divide into 2 portions.		
	To one portion, add KI(aq) followed by 1 cm ³ of starch solution.		brown solution formed (1) turns dark blue / blue–black on addition of starch. (1)
	To the other portion, add NaOH(aq).		brown ppt (1) insoluble in excess NaOH.
Scoring for QA of FA 9 and FA 10:			
Number of correct observations		Award (max 4)	
11 and more		4	
7 – 10		3	
4 – 6		2	
at least 3		1	

Qn	Suggested Mark Scheme	Mark															
	SUMMARY FA 9: Cu^{2+}, NO_3^- (do not accept NH_4^+, NO_2^-) FA 10: Fe^{2+} Correctly <u>identifies both cations</u> Correctly <u>identifies anion</u> and <u>supports answer with evidence</u> that NH_3 evolved with NaOH and Al powder (either NO_3^- or NO_2^-) but no NO_2 gas evolved on addition of acid (so can only be NO_3^-).	1 1															
(b)(i)	ligand exchange	1															
(ii)	redox	1															
(c)(i)	<table border="1"> <thead> <tr> <th></th><th>dilute HCl</th><th>NaOH(aq)</th></tr> </thead> <tbody> <tr> <td>MgO</td><td>soluble</td><td>insoluble</td></tr> <tr> <td>Al_2O_3</td><td>soluble</td><td>soluble</td></tr> <tr> <td>SiO_2</td><td>insoluble</td><td>insoluble</td></tr> <tr> <td>ZnO</td><td>soluble</td><td>soluble</td></tr> </tbody> </table> All correct At least 2 correct		dilute HCl	NaOH(aq)	MgO	soluble	insoluble	Al_2O_3	soluble	soluble	SiO_2	insoluble	insoluble	ZnO	soluble	soluble	2(max) 1
	dilute HCl	NaOH(aq)															
MgO	soluble	insoluble															
Al_2O_3	soluble	soluble															
SiO_2	insoluble	insoluble															
ZnO	soluble	soluble															
(ii)	Outline a logical sequence of testing that - successfully identifies FA 11 - reduces the possibility to 2 - reduces the possibility to 3 (Broad workable outline, not on details) Deduction with: - correct formula of species present (or balanced equations) - correct observations - explanations 8 out of 10 given ($\geq 80\%$) 5 – 7 ($\geq 50\%$) At least 3 (30%)	2(max) 1 0 3(max) 2 1															

Suggested Plan

Test	Observation	Deduction	Explanation
1. Add excess aq NaOH to a portion of FA 11 .	Dissolves (1)	Al ₂ O ₃ or ZnO	Both are <u>amphoteric oxides</u> , react with aq NaOH forming soluble salts: (1) $\text{Al}_2\text{O}_3(\text{s}) + 2\text{OH}^-(\text{aq}) + \text{H}_2\text{O}(\text{l}) \longrightarrow \underline{2\text{Al}(\text{OH})_4^-(\text{aq})}$ $\text{ZnO}(\text{s}) + 2\text{OH}^-(\text{aq}) + \text{H}_2\text{O}(\text{l}) \longrightarrow \underline{\text{Zn}(\text{OH})_4^{2-}(\text{aq})}$
	Does not dissolve (1)	SiO ₂ or MgO.	MgO is a <u>basic oxide</u> and has no reaction with a base/an alkali. (1) SiO ₂ an <u>acidic oxide</u> with giant molecular structure, no reaction with aqueous alkali nor soluble in aqueous solution. (1) (SiO ₂ would only react with hot conc. NaOH)
2. If oxide not soluble in step 1, add aq HCl to a portion of FA 11	Oxide insoluble (1)	SiO ₂	An acidic oxide which does not react nor dissolve in aq acid.
	Dissolves (1)	MgO	MgO is a basic oxide that reacts and dissolve in dil. HCl. $\text{MgO}(\text{s}) + 2\text{H}^+(\text{aq}) \longrightarrow \underline{\text{Mg}^{2+}(\text{aq})} + \text{H}_2\text{O}(\text{l})$
3. If oxide soluble in step 1, dissolve a portion of FA 11 in dil. HCl. To the solution, add aq NH ₃ till excess	white ppt, soluble in excess aq NH ₃ (1) white ppt, insoluble in excess aq NH ₃ (1)	ZnO Al ₂ O ₃	The amphoteric oxides react and dissolve in dil. HCl. $\text{ZnO}(\text{s}) + 2\text{H}^+(\text{aq}) \longrightarrow \text{Zn}^{2+}(\text{aq}) + \text{H}_2\text{O}(\text{l})$ $\text{Al}_2\text{O}_3(\text{s}) + 6\text{H}^+(\text{aq}) \longrightarrow 2\text{Al}^{3+}(\text{aq}) + 3\text{H}_2\text{O}$ $\text{Zn}^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq}) \longrightarrow \text{Zn}(\text{OH})_2(\text{s})$ $\text{Zn}(\text{OH})_2(\text{s}) + 4\text{NH}_3(\text{aq}) \longrightarrow \underline{\text{Zn}(\text{NH}_3)_4^{2+}(\text{aq})} + 2\text{OH}^-(\text{aq}) \text{ (1)}$ $\underline{\text{Al}(\text{OH})_3(\text{s})}$